

HARTFORD BRADLEY, CT, USA

WMO: 725080

Lat: 41.938N

Lon: 72.682W

Elev: 175

StdP: 14.60

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14740

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.9	9.1	-10.5	3.1	8.1	-5.8	4.1	12.5	27.2	27.2	24.5	25.1	7.2	320	0.519

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.2	91.6	73.4	88.7	71.9	85.8	70.4	76.4	86.6	74.9	83.9	73.4	81.4	10.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.4	124.8	80.5	72.1	119.6	79.1	70.8	114.3	78.0	39.8	86.6	38.4	84.2	37.0	81.7	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.8	19.4	17.4	-3.0	96.9	5.4	2.9	-6.9	99.0	-10.0	100.6	-13.1	102.3	-17.0	104.3
			-4.0	79.2	5.0	1.7	-7.7	80.4	-10.6	81.4	-13.4	82.4	-17.1	83.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.3	27.0	29.7	37.9	49.7	60.0	68.9	74.4	72.5	65.2	53.4	42.5	33.0
	DBStd	18.20	10.38	9.50	9.15	8.30	7.59	6.55	5.32	5.45	6.87	7.81	8.36	8.84
	HDD50	2617	713	570	390	102	6	0	0	0	0	53	252	531
	HDD65	5828	1177	989	841	465	197	34	2	6	82	369	674	992
	CDD50	3089	1	1	15	93	316	566	756	697	455	157	28	4
	CDD65	827	0	0	1	6	42	151	293	238	87	9	0	0
	CDH74	7400	0	0	12	100	511	1370	2697	1998	649	63	0	0
	CDH80	2559	0	0	2	33	170	463	1040	672	170	9	0	0
Wind	WSAvg	7.6	8.2	8.6	9.2	8.9	7.9	6.9	6.5	6.0	6.5	7.2	7.6	7.7
Precipitation	PrecAvg	45.8	3.3	3.2	3.7	3.7	4.0	4.1	3.8	3.9	4.2	3.9	3.8	3.9
	PrecMax	68.8	9.6	8.9	6.9	9.9	12.0	13.6	11.2	11.7	11.2	16.3	8.5	8.3
	PrecMin	29.4	0.4	0.4	0.3	1.1	0.9	0.7	1.0	0.6	0.8	0.4	0.4	0.8
	PrecStd	8.5	1.9	1.5	1.5	1.7	2.1	2.6	2.1	2.2	2.7	2.6	1.7	1.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.3	59.8	73.0	85.1	90.2	93.3	96.1	94.2	89.9	80.6	69.2	62.0
		MCWB	54.5	52.6	56.0	63.9	68.5	74.6	76.0	75.3	73.0	66.4	59.5	56.3
	2%	DB	51.2	52.7	63.1	75.4	84.9	89.1	92.0	89.5	84.8	74.6	64.3	55.4
		MCWB	46.9	45.1	50.6	57.3	66.7	72.0	73.9	73.5	69.8	64.1	57.6	51.3
	5%	DB	45.3	47.2	56.9	69.4	79.8	85.2	89.0	86.5	80.9	70.8	60.5	50.7
		MCWB	40.9	40.5	46.7	54.2	63.8	69.9	72.7	71.4	68.1	62.5	54.2	46.2
	10%	DB	40.5	42.4	51.6	64.4	74.8	81.9	85.8	83.7	77.4	66.7	56.4	45.9
		MCWB	36.4	37.2	44.0	51.6	60.8	67.5	70.9	69.8	66.2	58.7	50.0	41.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.1	54.9	59.5	65.5	72.3	76.8	78.5	78.0	75.2	70.8	64.0	57.5
		MCDB	57.8	57.3	69.2	79.5	86.1	88.9	90.5	89.8	83.5	75.7	66.6	59.1
	2%	WB	47.8	46.5	53.6	61.7	69.1	74.7	76.7	76.0	73.1	67.3	59.4	51.8
		MCDB	49.8	51.1	59.9	72.0	79.7	85.0	87.4	85.0	80.3	72.1	62.7	54.1
	5%	WB	41.3	41.2	48.5	57.7	66.4	72.7	75.2	74.4	71.2	63.8	55.3	46.7
		MCDB	44.7	45.9	55.1	64.6	75.7	81.3	84.4	82.1	77.3	69.1	59.4	50.3
	10%	WB	37.1	37.7	44.5	53.7	63.6	70.7	73.5	72.8	69.1	60.2	51.0	41.8
		MCDB	39.8	41.6	50.6	61.8	71.7	78.1	81.8	79.9	74.8	65.3	55.3	45.4
Mean Daily Temperature Range	5% DB	MDBR	15.1	16.4	17.8	20.7	21.3	20.3	20.2	20.1	20.5	19.6	17.0	14.5
		MCDBR	19.5	21.7	26.0	29.5	28.8	24.9	23.2	22.5	23.5	24.3	22.2	20.1
		MCWBR	16.4	15.8	16.0	15.1	13.7	11.4	8.9	9.0	11.1	14.1	16.1	16.9
	5% WB	MCDBR	18.9	20.0	23.3	24.2	24.7	21.1	19.8	19.0	18.6	20.4	19.4	19.2
	MCWBR	16.8	15.8	16.1	15.1	13.2	11.0	8.7	8.9	10.5	14.4	16.7	17.6	
Clear-Sky Solar Irradiance	taub	0.298	0.312	0.336	0.371	0.413	0.458	0.478	0.446	0.380	0.344	0.313	0.297	
	taud	2.435	2.411	2.424	2.366	2.282	2.187	2.142	2.220	2.381	2.496	2.538	2.524	
	Ebn at Noon	270	284	289	285	274	261	254	259	270	269	264	262	
	Edh at Noon	24	29	33	37	42	46	48	43	34	27	22	21	
All-Sky Solar Radiation	RadAvg	571	843	1163	1494	1665	1806	1868	1664	1367	896	626	468	
	RadStd	45	75	87	161	174	176	109	91	107	89	53	46	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+167	+103
Regional (84 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+180	+90

Nomenclature: See separate page